Aerospace series
Bolt, Blind, Protruding Head,
Self–Locking

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Contents

1 Scope

This standard specifies the required characteristics of a Blind, Protruding Head, Self Locking Bolt, for use in aerospace applications.

2 Normative references

This Airbus Standard incorporates by dated or undated reference, provisions from other publications. These normative references are cited at the appropriate places in the text and the publications are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus Standard only when incorporated in it by amendment or revision. For undated references the latest edition of the publication referred to applies (including amendments).

ABS2320	Fastener, Blind, High strength General technical specification
AMS2486	Conversion Coating of Titanium Alloys Fluoride-Phosphate Type
AMS2700	Passivation treatment for corrosion resistant steels
AMS4928	Titanium alloy bars and forgings
AMS4967	Titanium alloy bars and forgings
AMS5639	Steel corrosion resistant bars and forgings
AMS5731	Steel corrosion and heat resistant bars and forgings
AMS5732	Steel corrosion and heat resistant bars and forgings
AMS5737	Steel corrosion and heat resistant bars and forgings
AMS6930	Titanium Alloy Bars, Forgings and Forging Stock 6.0Al - 4.0V Solution Heat Treated and Aged UNS R56400
AMS-H-81200	Heat treatment of Titanium and Titanium alloys
AS5272	Lubricant, Solid Film, Heat Cured, Corrosion Inhibiting, Procurement Specification
AS87132	Lubricant, Cetyl Alcohol, 1-Hexadecanol, Application to Fasteners
ASTM D4181	Classification for Acetal moulding and extrusions
EN2424	Aerospace series - Marking of aerospace products
EN9100	Quality management systems - Requirements (based on ISO 9001:2000) and Quality systems - Model for quality assurance in design, development, production, installation and servicing
FCBF200	Procurement specification
ISO2768-1	General tolerances
MBF2000	Procurement Specification fasteners, blind, high strength for advanced composite materials
MBF2003	Installation and inspection specification

3 Requirements

3.1 Configuration, dimensions, tolerances and mass

3.1.1 The configuration, dimensions, tolerances and mass shall conform with figures 1 and 2 and tables 2 & 3. Tolerances not specified, shall be in accordance with ISO2768-1m.

3.1.2 Locking feature consists of (3) indentations located approximately 120° apart on the periphery of the nut component. Distortion of the shank shall not prevent insertion of the fastener into a ring gauge of diameter equivalent to minimum recommended hole size. Force of insertion shall not exceed 5.0 pounds.

3.1.3 Holes should be straight and perpendicular to surface, and should be reasonably round and free from delaminations.

3.1.4 Sheets should be firmly clamped together during drilling.

3.1.5 Edge of holes should be given a slight chamfer.

3.1.6 Core bolt break-off limits is measured from skin surface.

3.1.7 Installation and inspection specification to MBF2003

3.1.8 Mechanical properties shall be in accordance with table 4.

3.2 Material and surface treatment

Table 1: Material and surface treatment

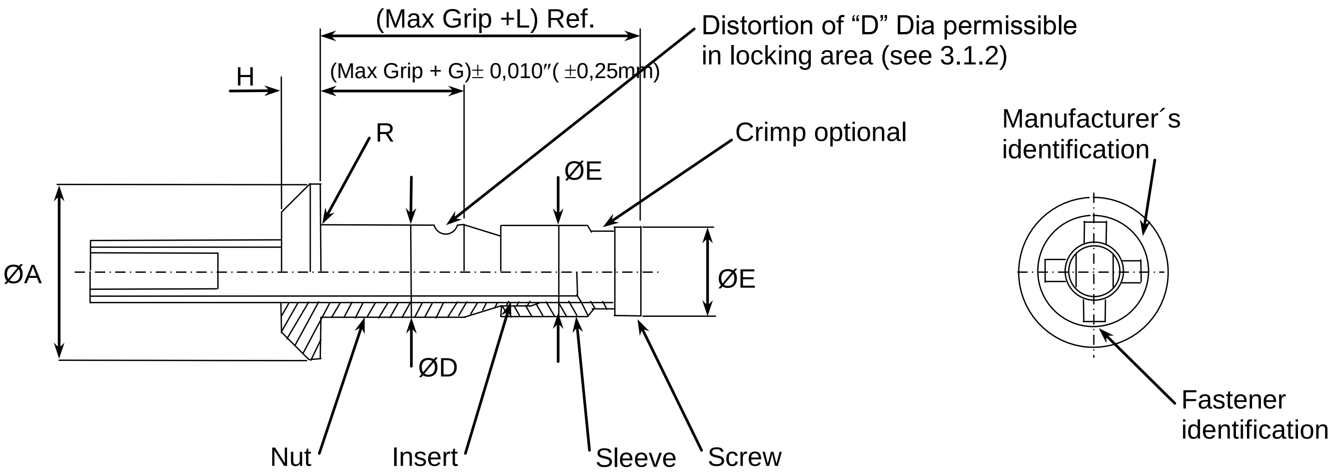
<table>
<tr><th>Item</th><th>Material</th><th>Heat Treatment</th><th>Surface treatment 1)</th><th>Code</th></tr>
<tr><td>Nut</td><td>6Al–4V Titanium
per AMS6930, AMS4928, AMS4967</td><td>AMS–H–81200
Maximum Hydrogen 125PPM</td><td>Phosphate Fluoride
as per AMS2486</td><td rowspan="5">None</td></tr>
<tr><td>Screw</td><td>A–286 with chemical composition
per AMS5731, AMS5732 or AMS5737</td><td rowspan="2">As Required for performance</td><td rowspan="2">Passivate per AMS2700 method 1 type 2, type 6, type 7 or type 8</td></tr>
<tr><td>Sleeve</td><td>304 Stainless steel
per AMS5639</td></tr>
<tr><td>Insert</td><td>ACETAL per ASTM D4181</td><td>–</td><td>None</td></tr>
<tr><td>(*) Drive Nut</td><td>Mild steel</td><td>As required for performance</td><td>Light grey corrosion protective coating</td></tr>
</table>

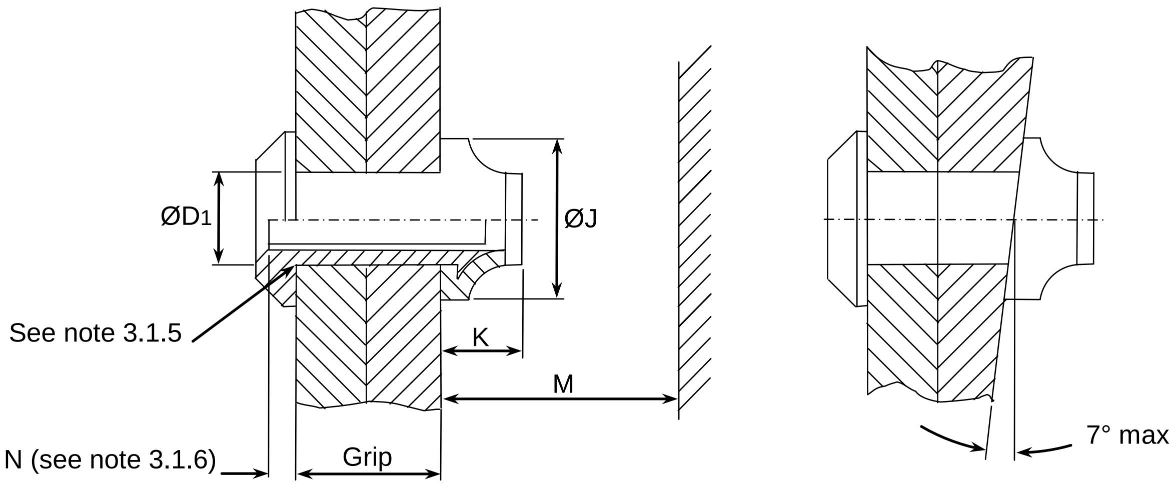
(*) code (Z) only

1) Note: Following lubricants may be applied to each one of the components by the fastener manufacturer, as required for performance (no other being allowed).

Nut: Cetyl Alcohol per AS87132.

Screw and Sleeve: Dry film lube per AS5272 Type I, Everlube 812, or Cetyl Alcohol per AS87132.

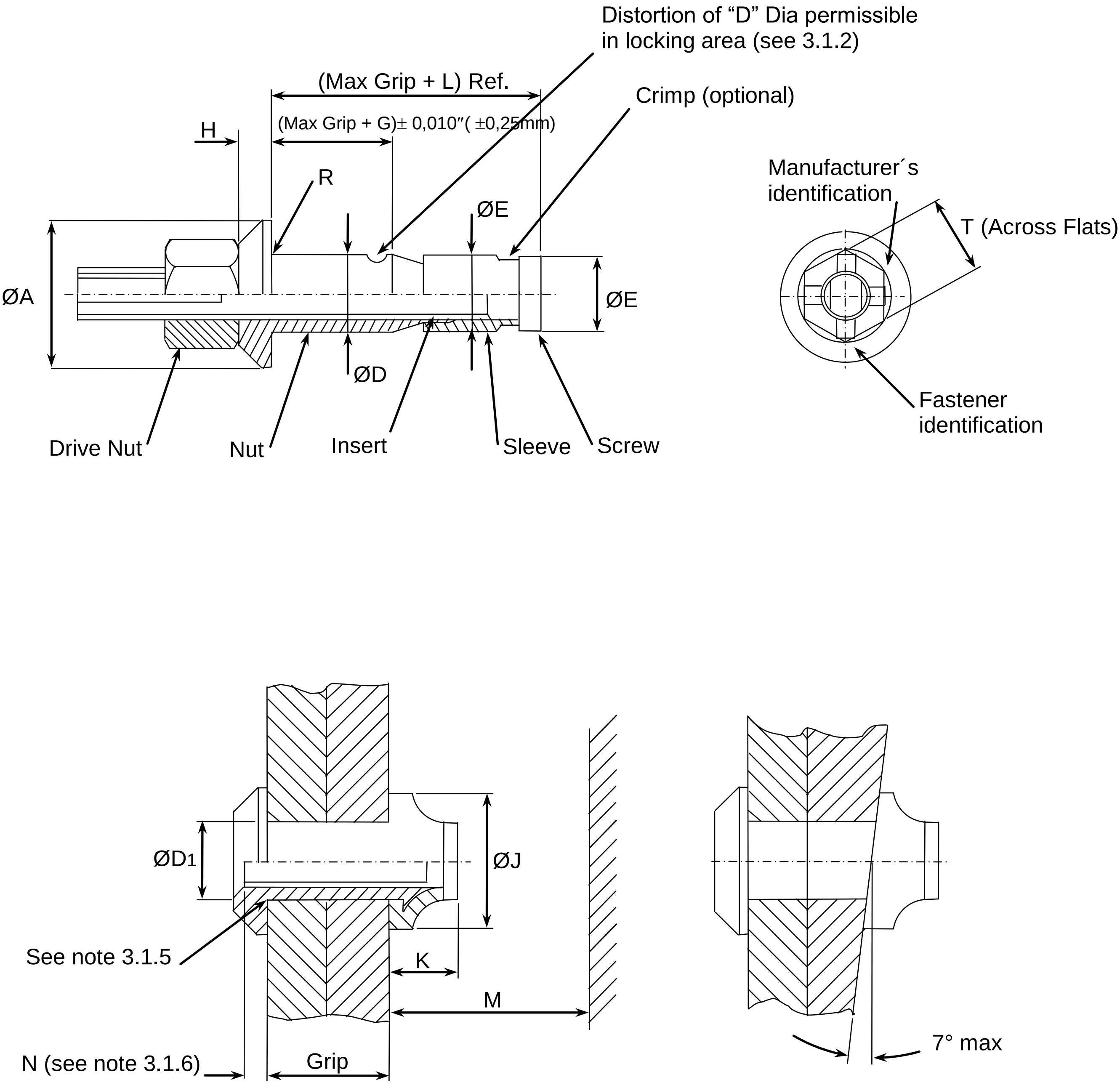




Typical installation
(see notes 3.1.3 and 3.1.4)

To be used on surfaces
with slope up to 7° max

Figure 1 – Code Y Configuration (Without drive nut for installation)



Typical installation
(see notes 3.1.3 and 3.1.4)

To be used on surfaces
with slope up to 7° max

Figure 2 – Code Z Configuration (With drive nut for installation)

Table 2 Dimensions and Tolerances

Dimensions in inch (mm)

<table>
<tr><th>Dia
Dash
Number</th><th>Nominal
Ø</th><th>ØA</th><th>ØD</th><th>ØE
Max</th><th>G
Ref</th><th>H</th><th>L
Ref</th><th>R
Rad
Max</th><th>T
Ref</th></tr>
<tr><td>–5</td><td>5/32
(4,0)</td><td>0.354
0.334
(8,99)
(8,48)</td><td>0.1645
0.1625
(4,178)
(4,128)</td><td>0.1640
(4,166)</td><td>0.017
(0,43)</td><td>0.060
0.053
(1,52)
(1,34)</td><td>0.512
(13,00)</td><td rowspan="4">0.030
(0,76)</td><td rowspan="4">0.375
(9,52)</td></tr>
<tr><td>–6</td><td>3/16
(4,8)</td><td rowspan="2">0.416
0.396
(10,57
(10,06)</td><td>0.1985
0.1965
(5,042)
(4,992)</td><td>0.1985
(5,042)</td><td>0.027
(0,69)</td><td rowspan="2">0.070
0.063
(1,78)
(1,60)</td><td>0.575
(14,61)</td></tr>
<tr><td>–7</td><td>7/32
(5,6)</td><td>0.2275
0.2255
(5,778)
(5,728)</td><td>0.2275
(5,778)</td><td>0.035
(0,89)</td><td>0.635
(16,13)</td></tr>
<tr><td>–8</td><td>1/4
(6,4)</td><td>0.541
0.521
(13,74)
(13,23)</td><td>0.2595
0.2575
(6,591)
(6,540)</td><td>0.2595
(6,591)</td><td>0.055
(1,40)</td><td>0.085
0.078
(2,16)
(1,98)</td><td>0.700
(17,78)</td></tr>
</table>

Table 2 Dimensions and tolerances (concluded)

Dimensions in inch (mm)

<table>
<tr><th colspan="7">Installation see notes 3.1.3 & 3.1.4</th></tr>
<tr><th>Dia
Dash
Number</th><th>Nominal
Ø</th><th>ØD1
Recm hole size
1)</th><th>ØJ
min</th><th>K
max</th><th>M
Ref</th><th>N
See note 3.1.6</th></tr>
<tr><td>–5</td><td>5/32
(4,0)</td><td>0.168
0.165
(4,267)
(4,192)</td><td>0.250
(6,35)</td><td>0.300
(7,62)</td><td>0.582
(14,78)</td><td rowspan="4">+0.103
-0.000
(+2,62)
(-0,00)</td></tr>
<tr><td>–6</td><td>3/16
(4,8)</td><td>0.202
0.199
(5,130)
(5,055)</td><td>0.300
(7,62)</td><td>0.350
(8,89)</td><td>0.645
(16,38)</td></tr>
<tr><td>–7</td><td>7/32
(5,6)</td><td>0.231
0.228
(5,867)
(5,792)</td><td>0.350
(8,89)</td><td>0.400
(10,16)</td><td>0.705
(17,91)</td></tr>
<tr><td>–8</td><td>1/4
(6,4)</td><td>0.263
0.260
(6,680)
(6,604)</td><td>0.400
(10,16)</td><td>0.450
(11,43)</td><td>0.770
(19,56)</td></tr>
</table>

1) For information only check applicable documentation

Table 3 Grip sizes and mass

Dia Dash Number					–5	–6	–7	–8
Nominal Diameter in mm					5/32 (4,0)	3/16 (4,8)	7/32 (5,6)	1/4 (6,4)
Grip Code No	Grip Range inch		mm		Mass kg/1000 Parts (Ref)			
	Max	Min	Max	Min				
–100	0.100	0.050	2,54	1,27	1,29	2,06	3,00	–
–150	0.150	0.100	3,81	2,54	1,38	2,20	3,18	4,59
–200	0.200	0.150	5,08	3,81	1,48	2,35	3,36	4,83
–250	0.250	0.200	6,35	5,08	1,57	2,49	3,54	5,07
–300	0.300	0.250	7,62	6,35	1,67	2,64	3,72	5,30
–350	0.350	0.300	8,89	7,62	1,76	2,79	3,90	5,53
–400	0.400	0.350	10,16	8,89	1,86	2,93	4,08	5,77
–450	0.450	0.400	11,43	10,16	1,96	3,07	4,26	6,00
–500	0.500	0.450	12,70	11,43	2,05	3,22	4,44	6,24
–550	0.550	0.500	13,97	12,70	2,15	3,36	4,62	6,47
–600	0.600	0.550	15,24	13,97	2,25	3,51	4,80	6,70
–650	0.650	0.600	16,51	15,24	2,34	3,65	4,98	6,94
–700	0.700	0.650	17,78	16,51	2,44	3,79	5,16	7,18
–750	0.750	0.700	19,05	17,78	2,53	3,94	5,34	7,41
–800	0.800	0.750	20,32	19,05	2,63	4,08	5,52	7,64
–850	0.850	0.800	21,59	20,32	2,72	4,23	5,70	7,88
–900	0.900	0.850	22,86	21,59	2,82	4,37	5,88	8,11
–950	0.950	0.900	24,13	22,86	2,91	4,51	6,06	8,35
–1000	1.000	0.950	25,40	24,13	3,01	4,66	6,24	8,58
–1050	1.050	1.000	26,67	25,40	3,10	4,80	6,42	8,81
–1100	1.100	1.050	27,94	26,67	3,20	4,95	6,60	9,05
–1150	1.150	1.100	29,21	27,94	3,29	5,09	6,78	9,28
–1200	1.200	1.150	30,48	29,21	3,39	5,24	6,96	9,52
–1250	1.250	1.200	31,75	30,48	3,48	5,38	7,14	9,75
–1300	1.300	1.250	33,02	31,75	3,58	5,53	7,32	9,99
–1350	1.350	1.300	34,29	33,02	3,80	5,78	7,56	10,76
–1400	1.400	1.350	35,56	34,29	3,90	5,95	7,76	11,01

Table 4 Mechanical Properties

Dia Dash No	Locking Torque (Min)		Double Shear Strength (Min)		Tensile Strength (Min)	
	In-lb	N-m	lb	daN	lb	daN
–5	1.0	0,11	3150	1401	900	400
–6	1.5	0,17	4600	2046	1400	623
–7	2.0	0,23	6050	2691	1600	712
–8	2.5	0,28	7900	3514	2100	934

4 Designation

Example :

Description block | Identity block

Bolt, Protruding Head | ABS0254–5–200 (Y)(Z)*

Number of this standard

Diameter dash number
(see table 2)

Grip code number (see table 3)

With or without drive nut

*Letters "Y" or "Z" are reserved for the sole use of manufacturing engineering and procurement Departments for selecting and ordering purposes.

Y = Fasteners without drive nut
Z = Fasteners with drive nut

5 Marking

EN 2424, style A

6 Technical specification

MBF 2000 or FCBF200 (depending on the supplier).
Or ABS2320 Type 1, Style A, Class 1, Code A, Subclass 1.

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
7 06/04	3.1.6	Measure change of the core bolt break off limit. Rewritten in new format.
8 11/04	3.2 (note)	Removal of paraffin wax lubricant.
9 11/14	2 3.17 Table 1 Figures 1 and 2 Table 2 4	Normative references updated 3.17 corrected to invoke MBF2003 table 1 updated to invoke correct specifications the word ‘Procurement’ removed from title Dimensions A, L and M corrected Wording for procurement amended to add manufacturing engineering
10 10/18	2 6	ABS0777 removed, ABS2320 added ABS2320 added